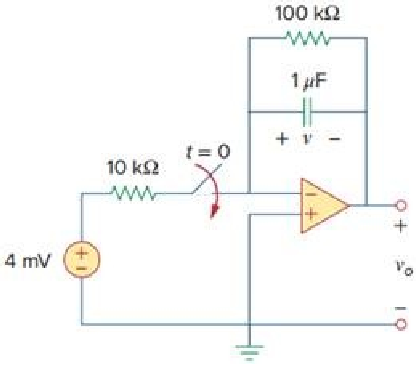
Chapter 7, Problem 15PP

Problem

Find $v(t)$ and $vo(t)$ in the op amp circuit of Fig. 7.58.

Fig. 7.58:



Step-by-step solution

Step 1 of 8

Refer to Figure 7.58 in the textbook.

Initially for $t<0$, the switch is opened for long time and the capacitor acts as an open circuit as shown in Figure 1.

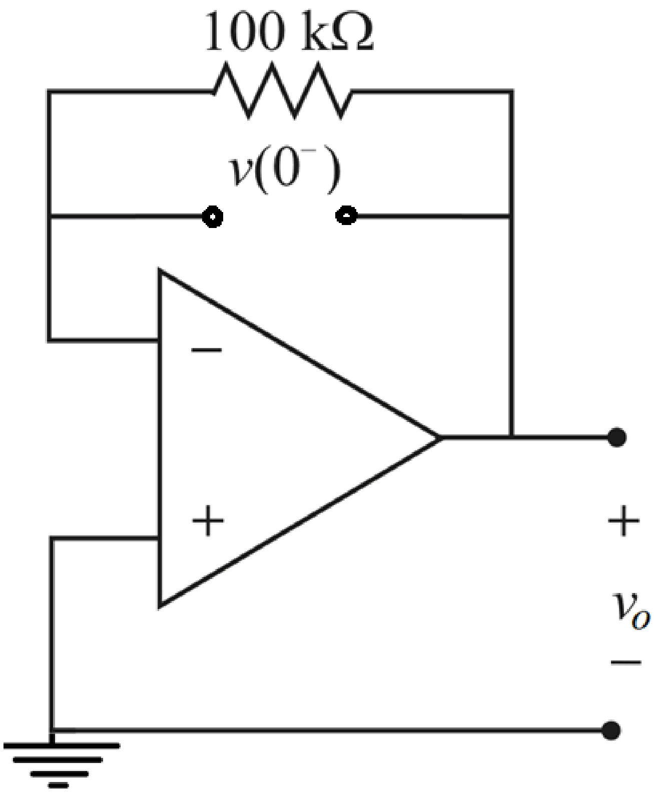


Figure 1

Step 2 of 8

Since it is a source free network, $v(0^-)=0$.

Capacitor voltage can't change instantaneously. Therefore,

$$\begin{aligned}v(0) &= v(0^-)\\ &= v(0^+)\\ &= 0\text{ V}\end{aligned}$$

Step 3 of 8

Draw the circuit diagram at $t=0$.

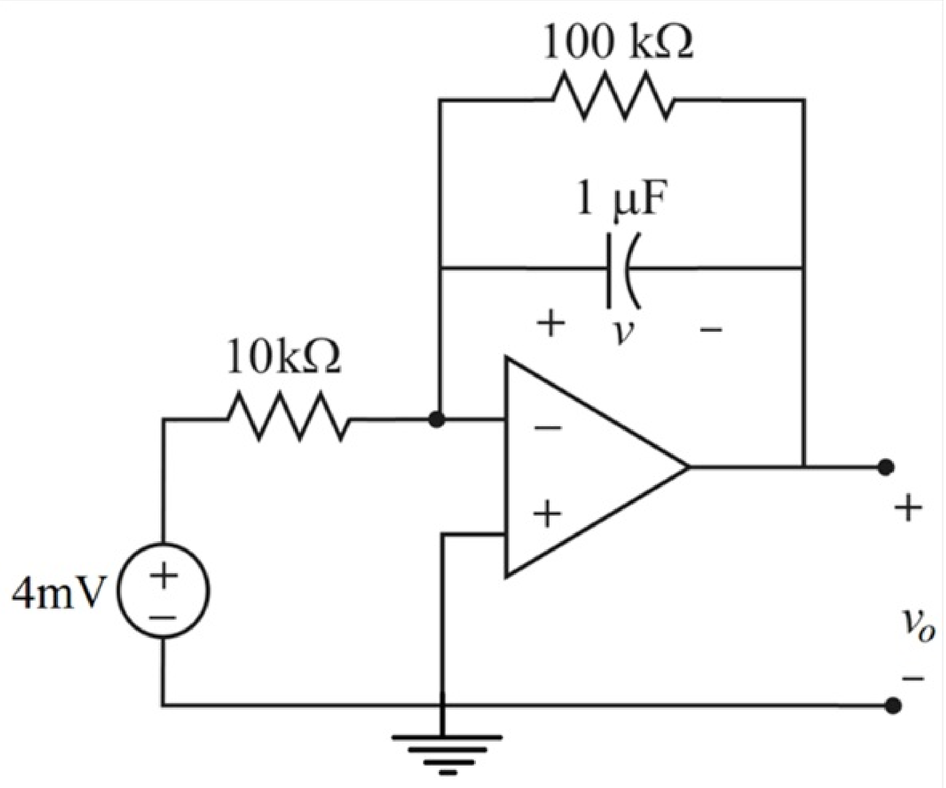


Figure 2

Step 4 of 8

From the virtual ground concept, the voltage at non-inverting terminal and inverting terminal is zero.

That is,

$$\begin{aligned}v_+ &= v_-\\ &= 0\text{ V}\end{aligned}$$

Apply Kirchhoff's current law at inverting node.

$$\frac{0-4\times10^{-3}}{10\times10^3}+\frac{v}{100\times10^3}+1\times10^{-6}\frac{dv}{dt}=0$$

$$\frac{-4\times10^{-3}}{10\times10^3}+\frac{v}{100\times10^3}+1\times10^{-6}\frac{dv}{dt}=0$$

$$-4\times10^{-7}+10^{-5}v+10^{-6}\frac{dv}{dt}=0$$

$$\frac{dv}{dt}+10\,v=0.4 \quad \text{...... (1)}$$

Step 5 of 8

The solution to the differential equation is,

$v(t) = A + Be^{-\frac{t}{\tau}}$ …… (2)

Differentiate the equation (2) with respect to t.

$$\frac{dv(t)}{dt} = \frac{d}{dt}\left[A + Be^{-\frac{t}{\tau}}\right]$$
$$\frac{dv(t)}{dt} = -\frac{B}{\tau}e^{-\frac{t}{\tau}}$$

Step 6 of 8

Substitute $-\frac{B}{\tau}e^{-\frac{t}{\tau}}$ for $\frac{dv(t)}{dt}$ and $A + Be^{-\frac{t}{\tau}}$ for $v(t)$ in equation (1).

$$-\frac{B}{\tau}e^{-\frac{t}{\tau}} + 10\left[A + Be^{-\frac{t}{\tau}}\right] = 0.4$$

$-\frac{B}{\tau}e^{-\frac{t}{\tau}} + 10A + 10Be^{-\frac{t}{\tau}} = 0.4$ …… (3)

Compare the constant terms in equation (3).

$$10A = 0.4$$
$$A = 0.04$$

Step 7 of 8

Substitute 0 for t in equation (1).

$$v(0) = A + Be^{-\frac{0}{\tau}}$$

$$0 = A + B$$

$$B = -0.04$$

Compare the exponential terms in equation (3).

$$10B - \frac{B}{\tau} = 0$$
$$10 = \frac{1}{\tau}$$
$$\tau = 0.1\text{sec}$$

Therefore, the expression for $v(t)$ is,

$$\begin{aligned} v(t) &= 0.04 - 0.04e^{-\frac{t}{0.1}}\text{ V} \\ &= 0.04\left(1 - e^{-10t}\right)\text{ V} \\ &= 40\left(1 - e^{-10t}\right)\text{ mV} \end{aligned}$$

Therefore, the expression for $v(t)$ is $\boxed{\begin{cases} 0\text{ V} & \text{for } t < 0 \\ 40\left(1 - e^{-10t}\right)\text{ mV} & \text{for } t > 0 \end{cases}}$.

Step 8 of 8

From Figure 2,

$$V_{-} - v_o = v$$
$$0 - v_o = v$$

$$\begin{aligned} v_o &= -v \\ &= -40\left(1 - e^{-10t}\right) \text{ mV} \\ &= 40\left(e^{-10t} - 1\right) \text{ mV} \end{aligned}$$

Therefore, the expression for $v_o(t)$ is $\boxed{\begin{cases} 0 \text{ V} & \text{for } t < 0 \\ 40\left(e^{-10t} - 1\right) \text{ mV} & \text{for } t > 0 \end{cases}}$.